

Saraswat Academy

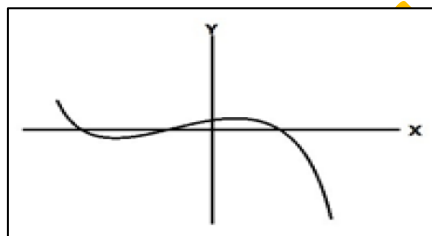
Class-X

Chapter-2 Polynomials

Class test

Multiple Choice Questions

Q.1 Find the number of zeroes of the given polynomial by its graph:



- (a) 2 (b) 3 (c) 1 (d) None of these

Q.2 If α & β are zeroes of the polynomial $3x^2 + 11x - 4$ then

$$\frac{1}{\alpha} + \frac{1}{\beta} \text{ equals}$$

- (a) $4/11$ (b) $-11/4$ (c) $11/4$ (d) $-4/11$

Q.3 If one zero of $2x^2 - 3x + k$ is reciprocal to the other then the value

of k is:

- (a) 3 (b) $2/3$ (c) $-3/2$ (d) 2

Q.4 A quadratic polynomial, whose zeroes are -3 and 4, is:

- (a) $x^2 - x + 12$ (b) $x^2 + x + 12$ (c) $x^2 - x - 12$ (d) $2x^2 + 2x - 24$

Q.5 If the zeroes of the quadratic polynomial $x^2 + (a + 1)x + b$ are 2 and -3, then:

- (a) $a = -7, b = -1$ (b) $a = 5, b = -1$ (c) $a = 2, b = -6$ (d) $a = 0, b = -6$

Q.6 If one zero of the quadratic polynomial $x^2 + 3x + k$ is 2, then the value of k is:

- (a) 10 (b) -10 (c) 5 (d) -5

Q.7 **Assertion (A):** $x^2 + 7x + 12$ has no real zeroes.

Reason (R): A quadratic polynomial can have at the most two zeroes.

- (a) Both assertion (A) and reason (R) are true and reason (R) is the correct explanation of assertion (A)
(b) Both assertion (A) and reason (R) are true and reason (R) is not the correct explanation of assertion (A)
(c) Assertion (A) is true but reason (R) is false.
(d) Assertion (A) is false but reason (R) is true.

Q.8 **Assertion (A):** If the sum of the zeroes of the quadratic polynomial $x^2 - 2kx + 8$ is 2 then value of k is 1.

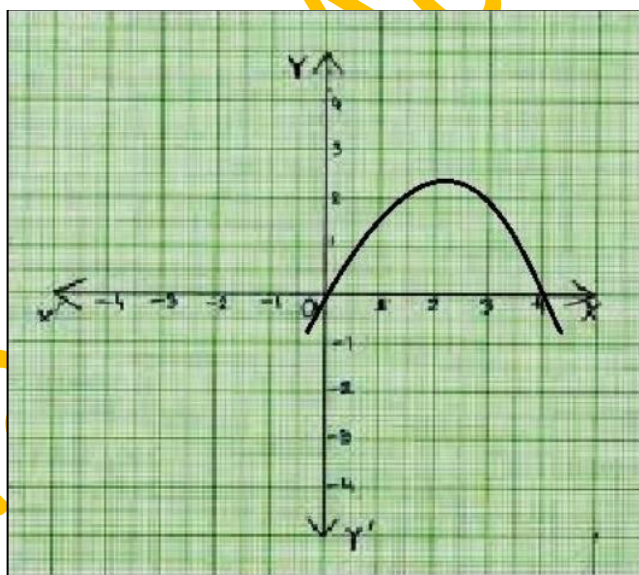
Reason (R): Sum of zeroes of a quadratic polynomial $ax^2 + bx + c$ is

$$-\frac{b}{a}$$

- (a) Both assertion (A) and reason (R) are true and reason (R) is the correct explanation of assertion (A)
- (b) Both assertion (A) and reason (R) are true and reason (R) is not the correct explanation of assertion (A)
- (c) Assertion (A) is true but reason (R) is false.
- (d) Assertion (A) is false but reason (R) is true.

Q.9 Case-Study 1

A boy is playing with a ball and suddenly he throws that ball upwards and ball returns to the ground describing a parabolic path as shown in the figure.



On the basis of above information, answer the following questions:

- (a) Find the zeroes of the given curve.

(b) Find the equation of the path traced by the ball.

(c) If $x = 0$, then find the value of the polynomial.

(d) If we shift the parabola to the right side by two units, then what will be its polynomial expression?

Q.10 Case-study 2

The below picture are few natural examples of parabolic shape which is represented by a quadratic polynomial. A parabolic arch is an arch in the shape of a parabola. In structures, their curve represents an efficient method of load, and so can be found in bridges and in architecture in a variety of forms.



On the basis of above information, answer the following:

- (a) In the standard form of quadratic polynomial $ax^2 + bx + c$; a, b & c are?
- (b) If α & $1/\alpha$ are zeroes of $2x^2 - x + 8k$, then find the value of k .
- (c) What type of graph does the polynomial $x^2 + 1 = 0$ forms?
- (d) If the sum of zeroes is $-p$ and the product of zeroes is $-1/p$, then form the quadratic polynomial.